4- Inside a computer system (This room covers the basic components of computer system) :-

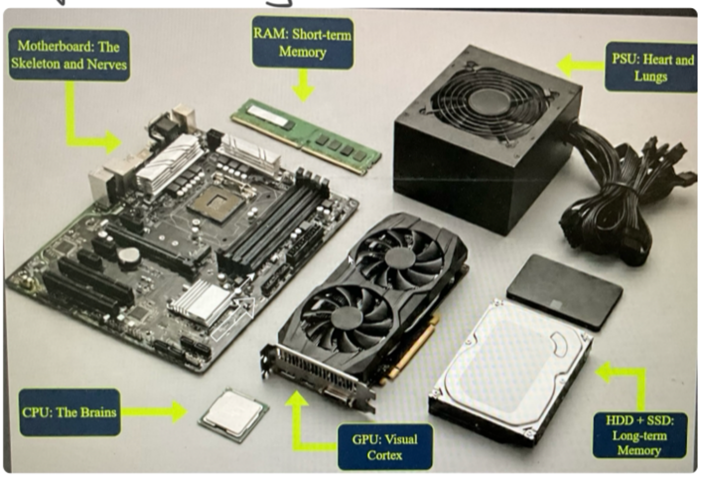
welcome to first room of Computer Fundamentals module ! before we need to talk about security, we need to first understand what we are securing.

learning objectives :-

* After completing this room, you will be able to recognize and understand functions of various computing components

Task 2 :- Inside a computer System

Nearly every computer System that you can think of Includes, In one way or another, the same building blocks. Each part has its own job, and together they make computer work. lets look at each of these building blocks.



To learn more about open view site.

Motherboard

Q) what role does the Motherboard play in a computer?

A) It processes all calculations and Instructions

B) It connects and allows communication between all components

C) It stores data permanently even when powered off.

Motherboard is like our bodys skeleton and nervous System. It holds all different components in place and connects them. Typical desktop motherboard, you'll see different connectors that house all your components - CPU socket, RAM slots, expansion slots and various ports.

Every other component plugs into or connects through motherboard.

CPU :- (Central processing unit)

Q) What do you think CPU does?
A) Stores files and programs permanently
B) Executes Instructions and perform calculations
C) Converts Electricity for other components.

CPU (Central processing unit), often called processor, is comparable to part of our brain. Just like our brain continuously executes Instructions (add numbers, pour milk in bowl, and so on) a CPU does same for computer. Modern CPUs have multiple cores that handles Instructions in parallel.

The CPU connects to motherboard via CPU socket.

Q) What advantage do modern CPUs have multiple cores?

A) They can store more data
B) They can process multiple instructions at same time.
C) They use less electricity

RAM (Random Access Memory) :-

Q) How would you describe what RAM does?
A) Stores data temporarily while computer is running
B) Saves files permanently like photos and documents
C) Displays images on the monitor.

RAM is comparable to our brains short term or working memory. When working on task, we keep relevant information in mind temporarily. RAM does same it holds data that CPU needs quick access to.

RAM is Volatile :- when power is lost, all contents are gone. Modern RAM modules use technologies like DDR5 or DDR6 for increased speed and performance.

Storage (SSD/HDD) :-

Q) What's the main purpose of storage devices like SSDs or HDDs?
A) To process instructions faster than the CPU

B) To hold data temporarily while programs run
C) To save data permanently even when powered off

SSDs and HDDs are storage devices, comparable to our long term memory. Just like fond memories are remembered permanently. Data is saved long term on storage devices.

HDDs use older technology with moving parts, limiting performance. SSDs have no moving parts and use memory chips, allowing much faster speeds.

Network Adapter: —

Q) what does a network adapter enable a computer to do?
A) Store more files in the cloud
B) communicate with other computers and networks
C) Run Programs faster

Just like you use vocal cords to Communicate with our environment, a network adapter lets computers communicate with other systems. A Network Adapter Comes in wireless and wired variants.

Power supply (PSU): —

Q) what is the main function of Power Supply unit?

A) To cool down the computer components
B) To supply electrical power to all components
C) To boost computers processing Speed.

Every System needs power. just as our heart pumps blood to our organs, a PSU supplies energy to all system components. If components need more power than PSU can provide, the system will fail.

Graphics Cards: —

Q) what does a graphics card primarily handle?
A) processing and outputting visual information to display
B) Connecting to Internet faster
C) storing game files and videos

The Graphic card is comparable to visual cortex of our brain. Our eyes pick up information and visual cortex processes it into Images. Similarly graphic card receives information from operating system and programs, then outputs processed visual data to monitor.

Input/output: —

Q) what are input and output devices used for?
A) only for gaming and entertainment
B) To send data and receive data from computer

C) To increase the computers memory

Input devices: — Keyboards, microphones, mice and scanners. Output devices include monitors, printers and speakers.